

PS 160 — Physics II – Final Exam

Instructor name: _____ Section: _____

Date: Sat 27 Apr 2024

Student name _____

Score: /250

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- RELAX.
 - By starting this exam, you commit yourself to Embry-Riddle Aeronautical University's Code of Conduct.
 - You are allowed the instructor-provided equation sheet and a hand-held calculator. Aid from other sources (internet, other people, etc) is not allowed.
 - Look over all questions before you begin, and attempt the easier ones first. Place your answers in the boxes provided.
 - There will be no partial credit. To receive full credit you must SHOW your work. Your handwritten work MUST be submitted in a clear, concise manner.
 - You will be graded on your best 250 out of a total of 300 points.
 - All problems are worth 10 points unless otherwise noted.
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1. Calculate the pressure (in Pa) at the bottom of a cylinder of mercury 46 cm deep if the ambient atmospheric pressure is 71 kPa. ($\rho_{Hg} = 13.6 \text{ g/cm}^3$)

$p =$ Pa

2. A hydraulic jack has an input piston with cross-sectional area 0.03 m^2 and an output piston with cross-sectional area 0.25 m^2 . If the input force is 640 N , calculate the output force in kN .

$$F = \boxed{} \text{ kN}$$

3. A simple pendulum has length 1.87 m and the bob has mass 4.1 kg . If the local acceleration due to gravity is 7.3 m/s^2 (we're on an exoplanet), calculate the period of small oscillations of the pendulum.

$$T = \boxed{} \text{ s}$$

4. A 3.4-kg block attached to a spring of force constant 288 N/m undergoes simple harmonic motion of amplitude 0.14 m . What is the energy of the mass-spring system?

$$E = \boxed{} \text{ J}$$

5. A vibrating string 55 cm long with mass 3.0 g is under a tension of 12 N. It vibrates in the $n = 3$ normal mode. Calculate (a) the frequency, and (b) the wave speed of the traveling waves on this string.

(a) $f =$ Hz

(b) $v =$ m/s

6. [20 points] Suppose a wave function is given by $y(x, t) = 5 \cos(3t + 0.5x - 2.1)$ where all values are in MKS units. Then ...

(a) the amplitude is m

(b) the angular frequency is rad/s

(c) the wavenumber is rad/m

(d) the phase shift is rad

(e) the frequency is Hz

(f) the period is s

(g) the wavelength is m

(h) the propagation speed of the wave is m/s

(i) the transverse **speed** of the wave at $(x = 0, t = 0)$ is m/s

7. A siren is designed to create a *sound intensity level* of 116 dB at a distance of 6 m from the siren. Calculate the *sound intensity* in mW/m^2 at a distance of 39 m from the siren.

$$I = \boxed{} \text{ mW}/\text{m}^2$$

8. A motorcyclist is traveling at 49 m/s away from a police car at rest. If the policeman blows his horn at a frequency of 439 Hz, what frequency in Hz does the motorcyclist hear? The speed of sound is 340 m/s.

$$f = \boxed{} \text{ Hz}$$

9. Granite has a coefficient of linear expansion of $3.5 \times 10^{-5} \text{ K}^{-1}$. Suppose a sphere made of granite has a volume of 1 m^3 . Calculate the temperature change (in $^\circ\text{C}$) if the volume increases by 10^{-3} m^3 .

$$\Delta T = \boxed{} ^\circ\text{C}$$

10. A 9-kg cube of iron at 90 °C falls into 2 kg of cold water at 10 °C. Neglecting the container and any energy loss, calculate the final equilibrium temperature of the iron and water. [$c_{iron} = 444$ J/(kg·K) and $c_{water} = 4190$ J/(kg·K)]

$$T = \boxed{} \text{ } ^\circ\text{C}$$

11. An ideal gas increases in temperature from 220°C to 450°C, and at the same time the volume decreases by 50%. If the initial pressure is 89 kPa, what is the final pressure?

$$p = \boxed{} \text{ kPa}$$

12. What is the total translational kinetic energy of 7.2 mol of an ideal gas at 23°C?

$$K_{tr} = \boxed{} \text{ } ______ \text{ (insert units here)}$$

13. An ideal gas decreases in volume from 1.8 m^3 to 1.2 m^3 at a constant pressure of 50 kPa . Calculate the work done by the gas during this process.

$$W = \boxed{} \text{ kJ}$$

14. A diatomic ideal gas undergoes an isothermal expansion starting from a volume of 1.3 m^3 and pressure 53 kPa . If the final volume is 2.7 m^3 , calculate the change in internal energy.

$$\Delta U = \boxed{} \text{ (insert units here)}$$

15. **[20 points]** An engine operates in a Carnot cycle between two reservoirs of temperature 305 K and 799 K . (a) How much work does the engine do in one cycle if it also absorbs 2400 J from the high temperature reservoir? (b) What is the efficiency of this engine?

$$(a) W = \boxed{} \text{ J}$$

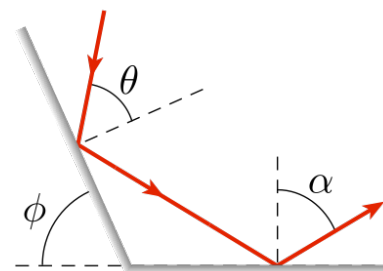
$$(b) e = \boxed{}$$

16. Suppose 11 kg of water at 0 °C freezes to ice at 0 °C. Calculate the entropy change of this process (in kJ/K). Carefully consider the sign of ΔS . (The latent heat of fusion of water is 3.35×10^5 J/kg.)

$$\Delta S = \boxed{} \text{ kJ/K}$$

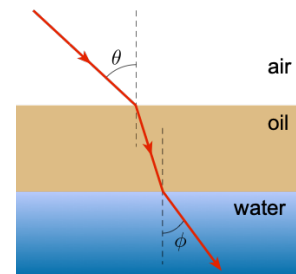
17. The figure shows two plane mirrors that meet at a corner. A ray of light is incident on the first mirror with an angle of incidence equal to θ . Calculate the angle α between the outgoing ray and the second mirror's normal if $\phi = 60^\circ$ and $\theta = 60^\circ$.

$$\alpha = \boxed{} \text{ degrees}$$



18. The figure shows a light ray incident on a layer of oil (at an angle of incidence θ) that is floating on water. Calculate the final angle of refraction ϕ in degrees if $n_{oil} = 1.59$, $n_{water} = 1.33$, and $\theta = 33^\circ$.

$$\phi = \boxed{} \text{ degrees}$$



19. **[20 points]** An object is located 18 cm in front of a concave spherical mirror of focal length 8 cm. Determine the following:

(a) the image position, $s' =$ cm

(b) the lateral magnification, $m =$

(c) the image is (real/virtual)

(d) the image is (upright/inverted)

20. **[20 points]** An object is located 16 cm in front of a diverging lens of focal length 12 cm. Determine the following:

(a) the image position, $s' =$ cm

(b) the lateral magnification, $m =$

(c) the image is (real/virtual)

(d) the image is (upright/inverted)

21. A liquid-air boundary has a critical angle of 43.5° for total internal reflection. If a ray of light traveling in air has an angle of incidence of 28° at the liquid-air boundary, calculate the angle between the refracted ray and the surface normal.

$$\theta_{refraction} = \boxed{} \text{ degrees}$$

22. Light reflecting off a bubble in the shape of a spherical shell exhibits enhancement in reflected light at a wavelength of 471 nm. Calculate the bubble's minimum thickness (in nm) if its index of refraction is 1.5.

$$t = \boxed{} \text{ nm}$$

23. A single-slit experiment involves monochromatic, coherent light incident on a rectangular aperture of width 0.31 mm. If the screen is 2 m away and the distance from the central maximum to the $m = 2$ minimum is 7.3 mm, determine the wavelength of the light (in nm).

$$\lambda = \boxed{} \text{ nm}$$

24. Monochromatic, coherent light of wavelength 418 nm is incident on two slits separated by 0.26 mm. If the screen is 2.1 m away, calculate the full width (in mm) of the central maximum on the screen.

$$\Delta y = \boxed{} \text{ mm}$$

25. Suppose a space telescope orbiting Mars has an aperture diameter of 4.5 cm. What minimum width crater (in meters) can it resolve in light of wavelength 557 nm if it is in an orbit 883 km from the planet's surface? Disregard atmospheric effects.

$$d_{min} = \boxed{} \text{ m}$$

26. Light of wavelength 0.54 nm creates a first-order interference maximum when it is incident on a crystal at a grazing angle of 23° from the surface. Determine the interatomic spacing of the Bragg planes of the crystal (in nm).

$$d = \boxed{} \text{ nm}$$